

SMART CONTRACTS REVIEW

zokyo

October 22nd 2024 | v. 1.0

Security Audit Score

PASS

Zokyo Security has concluded that these smart contracts passed a security audit.



ZOKYO AUDIT SCORING SYMBIOSIS

1. Severity of Issues:
    - Critical: Direct, immediate risks to funds or the integrity of the contract. Typically, these would have a very high weight.
    - High: Important issues that can compromise the contract in certain scenarios.
    - Medium: Issues that might not pose immediate threats but represent significant deviations from best practices.
    - Low: Smaller issues that might not pose security risks but are still noteworthy.
    - Informational: Generally, observations or suggestions that don't point to vulnerabilities but can be improvements or best practices.
2. Test Coverage: The percentage of the codebase that's covered by tests. High test coverage often suggests thorough testing practices and can increase the score.
3. Code Quality: This is more subjective, but contracts that follow best practices, are well-commented, and show good organization might receive higher scores.
4. Documentation: Comprehensive and clear documentation might improve the score, as it shows thoroughness.
5. Consistency: Consistency in coding patterns, naming, etc., can also factor into the score.
6. Response to Identified Issues: Some audits might consider how quickly and effectively the team responds to identified issues.

SCORING CALCULATION:

Let's assume each issue has a weight:
- Critical: -30 points
- High: -20 points
- Medium: -10 points
- Low: -5 points
- Informational: -1 point

Starting with a perfect score of 100:
- 0 Critical issues: 0 points deducted
- 0 High issues: 0 points deducted
- 0 Medium issues: 0 points deducted
- 1 Low issue: 1 resolved = 0 points deducted
- 3 Informational issues: 3 resolved = 0 points deducted

Thus, the score is 100

TECHNICAL SUMMARY

This document outlines the overall security of the Symbiosis smart contract/s evaluated by the Zokyo Security team.

The scope of this audit was to analyze and document the Symbiosis smart contract/s codebase for quality, security, and correctness.

Contract Status



There were 0 critical issues found during the review. (See Complete Analysis)

It should be noted that this audit is not an endorsement of the reliability or effectiveness of the contract/s but rather limited to an assessment of the logic and implementation. In order to ensure a secure contract that can withstand the TON network's fast-paced and rapidly changing environment, we recommend that the Symbiosis team put in place a bug bounty program to encourage further active analysis of the smart contract/s.

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AUDITING STRATEGY AND TECHNIQUES APPLIED

The source code of the smart contract was taken from the Symbiosis repository:
Repo: Zip File: ton-contracts-master.zip
Sha256: 04f75639b7423aa947ae3fefb84f0cba2a110369729304c59b5048b5caf8a5a7

The repository with the last commit: https://github.com/symbiosis-finance/ton-contracts-public/commit/7999a18d482f443d73ed3913aa5368a380390a18

Contracts under the scope:

- ./contracts/bridge.fc 962d014aa5ce1cbc96978cd13bcc31f887e2205c0a01502bd466e3e6fc2a4030
- ./contracts/jetton/jetton-utils.fc f825c69d4660f04267b2b0f2372d65819ac9c90b7bf828910bfc16a1bdcd9b19
- ./contracts/jetton/params.fc 5c544ab1d36a05d60649389a24cfee6a3489f49ad5790fe430b890e3760f2c79
- ./contracts/jetton/jetton-minter.fc 4fc34d4a0dd8cd8453dcb3d5f1ff261346713792a6cab23783aecbef1a196530
- ./contracts/jetton/jetton-wallet.fc 9369b73c1938584ea65061f0d3a13043d2c32b07c8e217576fdbbff953057a22
- ./contracts/jetton/op-codes.fc 803967bb1ab82be89ca4d9e530cd5f8b347426ab3c101c24c7377d09cfd543e0
- ./contracts/imports/stdlib.fc 1c510f50d92d2a8d012a96baea1af06a2aa00e2c169f2e3f9d65a95f94c24131
- ./contracts/external_id.fc ef0a7e14915526af618b32f65d8d8f2600e688a8d3211a0d6a683d22f0663e90
- ./contracts/utils.fc 505f246b1450e3fd78eaa2df39f195cb60d56fc6b28e98a5d355c7d0a5a22c8c

During the audit, Zokyo Security ensured that the contract:

- Implements and adheres to the existing standards appropriately and effectively;
- The documentation and code comments match the logic and behavior;
- Distributes tokens in a manner that matches calculations;
- Follows best practices, efficiently using resources without unnecessary waste;
- Uses methods safe from reentrance attacks;
- Is not affected by the most recent vulnerabilities;
- Meets best practices in code readability, etc.

Zokyo Security has followed best practices and industry-standard techniques to verify the implementation of Symbiosis smart contract/s. To do so, the code was reviewed line by line by our smart contract developers, who documented even minor issues as they were discovered. In summary, our strategies consist largely of manual collaboration between multiple team members at each stage of the review:

	Due diligence in assessing the overall code quality of the codebase.
	Thorough manual review of the codebase line by line.
	Cross-comparison with other, similar smart contract/s by industry leaders.

Executive Summary

STRUCTURE AND ORGANIZATION OF THE DOCUMENT

For the ease of navigation, the following sections are arranged from the most to the least critical ones. Issues are tagged as “Resolved” or “Unresolved” or “Acknowledged” depending on whether they have been fixed or addressed. Acknowledged means that the issue was sent to the Symbiosis team and the Symbiosis team is aware of it, but they have chosen to not solve it. The issues that are tagged as “Verified” contain unclear or suspicious functionality that either needs explanation from the Client or remains disregarded by the Client. Furthermore, the severity of each issue is written as assessed by the risk of exploitation or other unexpected or otherwise unsafe behavior:

Critical

The issue affects the contract in such a way that funds may be lost, allocated incorrectly, or otherwise result in a significant loss.

High

The issue affects the ability of the contract to compile or operate in a significant way.

Medium

The issue affects the ability of the contract to operate in a way that doesn’t significantly hinder its behavior.

Low

The issue has minimal impact on the contract’s ability to operate.

Informational

The issue has no impact on the contract’s ability to operate.

COMPLETE ANALYSIS

FINDINGS SUMMARY

#	Title	Risk	Status
1	Unused variables	Low	Resolved
2	Unused code	Informational	Resolved
3	Unused variable	Informational	Resolved
4	Unused function argument	Informational	Resolved

LOW-1 | RESOLVED

Unused variables

Where:
- bridge.fc:258
- bridge.fc:262
Description: there are two variables loaded from the `inner_payload_slice1` but never used afterward.

Recommendation:
ensure you don't need to verify these values and remove variable declarations.

INFORMATIONAL-1 | RESOLVED

Unused code

Where:
- utils.fc:4
Description: there is a function `dump_slice` declared in the file, which is never used in the code

Recommendation:
do not keep debug code in the production codebase, remove it

INFORMATIONAL-2 | RESOLVED

Unused variable

Where:
- bridge.fc:203
- utils.fc:105
- external_id.fc:25

Description: there are variables declared that are never used in the code

Recommendation:
remove the variable declarations

INFORMATIONAL-3 | RESOLVED

Unused function argument

Where:
- bridge.fc:321
- bridge.fc:439
- bridge.fc:520
- bridge.fc:578
- external_id.fc:30

Description: function declarations contain arguments that aren't used in the code

Recommendation:
remove the variable declarations

• **./contracts/bridge.fc** • **./contracts/jetton/jetton-utils.fc** • **./contracts/jetton/params.fc** • **./contracts/jetton/jetton-minter.fc** • **./contracts/jetton/jetton-wallet.fc** • **./contracts/jetton/op-codes.fc** • **./contracts/imports/stdlib.fc** • **./contracts/external_id.fc** • **./contracts/utils.fc**	
Re-entrancy	Pass
Access Management Hierarchy	Pass
Arithmetic Over/Under Flows	Pass
Unexpected Ether	Pass
Delegatecall	Pass
Default Public Visibility	Pass
Hidden Malicious Code	Pass
Entropy Illusion (Lack of Randomness)	Pass
External Contract Referencing	Pass
Short Address/ Parameter Attack	Pass
Unchecked CALL Return Values	Pass
Race Conditions / Front Running	Pass
General Denial Of Service (DOS)	Pass
Uninitialized Storage Pointers	Pass
Floating Points and Precision	Pass
Tx.Origin Authentication	Pass
Signatures Replay	Pass
Pool Asset Security (backdoors in the underlying TRC-20)	Pass

We are grateful for the opportunity to work with the Symbiosis team.

The statements made in this document should not be interpreted as an investment or legal advice, nor should its authors be held accountable for the decisions made based on them.

Zokyo Security recommends the Symbiosis team put in place a bug bounty program to encourage further analysis of the smart contract by third parties.



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